Airline Operations Desk: A Briefing Guide

This overview provides the foundational knowledge needed to understand airline operations.

The Role of Operations Control

An airline monitors its daily schedule through an Operations Control Centre (OCC). Dispatchers, duty managers, and hub coordinators are responsible for keeping the schedule intact when disruptions occur. They are neither pilots nor passengers; instead, they make critical decisions regarding aircraft routing, crew assignments, and passenger connections hours ahead of departure.

The Cascade Effect of Delays

Aircraft fly multiple consecutive legs (e.g., A to B to C to D). A late arrival at B typically causes a late departure for C, creating a ripple effect. Prolonged delays can cause crews to exceed legal duty limits, forcing flight cancellations, and causing passengers to miss hub connections. The most significant financial impact comes from this cascading effect, making early intervention the controller's primary objective.

Proactive Mitigation

Controllers manage disruptions through calculated actions, such as:

- Holding a connecting flight to accommodate delayed passengers.
- Swapping aircraft assignments.
- Deploying standby crews.
- Adjusting departure times to avoid airspace congestion.

Because every operational change incurs a cost, controllers must accurately assess whether a risk justifies the intervention.

Primary Causes of Delay

Identifying the root cause dictates the response strategy. Delays generally fall into four standard categories:

- **Carrier:** Internal airline issues, such as slow turnarounds or mechanical faults.
- **NAS (National Airspace System):** Congestion at the airport or within the air traffic control system.
- **Weather:** Broad disruptions impacting operations at a specific location.
- **Late Aircraft:** The compounding effect of a previously delayed inbound flight.

The Current Tool Gap

Today's controllers rely heavily on personal experience and fragmented systems, monitoring separate screens for schedules, weather, and live radar. There is a lack of integrated tools capable of fusing these data points to identify which flights are at risk of slipping and explaining exactly why. Bridging this gap is your primary challenge.

Requirements for a Trusted Tool

Operations controllers will not act on a raw probability score. A generic alert like "Flight LH123 is high risk" will be ignored. To be trusted and utilized, a tool must provide the underlying context and a recommended action: "LH123 is high risk due to a 40-minute late inbound aircraft and a tight turnaround; consider an aircraft swap." The reasoning and the recommendation are the actual product.

Project Scope and Constraints

For this weekend's sprint:

- **Out of scope:** Interacting with a live airline, sourcing private data, or learning aviation regulations.
- **In scope:** Build on the provided dataset, clearly define your target user (e.g., dispatcher, hub coordinator), and focus your energy on the core challenge of translating a calculated risk into a clear, actionable decision.